

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457282.7493 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.205 +/- 0.063
Transit depth (Rp/Rs)^2	4.22 +/- 2.57 [%]
Orbital Inclination (inc)	84.66 +/- 2.82
Airmass coefficient 1 (a1)	0.73 +/- 0.47
Airmass coefficient 2 (a2)	0.17 +/- 0.81
Scatter in the residuals of the lightcurve fit is	56.28 %
Best Comparison Star	None
Optimal Aperture	3.03
Optimal Annulus	6.59
Transit Duration (day)	0.066 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.205 ± 0.063 vs 0.1646 (deviation 0.64σ)
Duration fit vs published	0.066 d vs 0.0758 d (deviation 0.36σ)
Mid-time O–C	5.4 min
Depth SNR	3.3
Residual scatter	56.28 %

Light curve

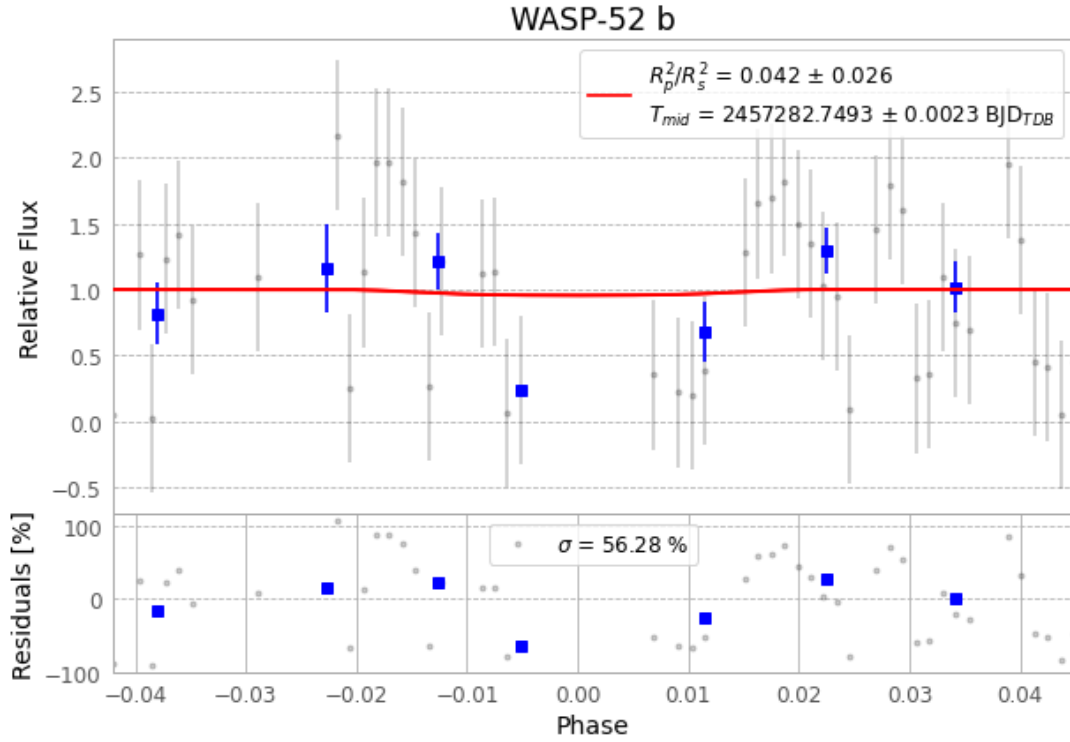


Figure 1 — FinalLightCurve_WASP-52 b_2015-09-16.png (SHA-256 6129914641b9f401...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [513, 301], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 58afe52adf33ec08...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-52150917040012.FITS → WASP-52150917074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-150917.log (c69375fcd31e...), FinalLightCurve_WASP-52 b_2015-09-16.png (6129914641b9...), FinalLightCurve_WASP-52 b_2015-09-16.csv (549a1bbe4e15...)

Compute resources

Wall time 295.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:15Z.